

Hiver SDE Intern Take-Home

Autonomous AI Customer Support Agent • American Airlines
(@AmericanAir)

DELIVERABLE #5

Engineering & Science Decision Log

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Core Assignment Philosophy: *"The proof is worth more than the system."*

This log documents the **15 non-obvious engineering and scientific decisions** made while converting a messy, noisy Kaggle Twitter dataset (~3M tweets) into a production-grade customer support agent. Every decision is backed by empirical numbers, real failure analysis, and deliberately rejected alternatives.

76.0%

SOTA INTENT
ACCURACY (+18%
GAIN)

5.0 / 5.0

GROUND
ED
RESOLUTION SCORE

r = 0.628

HUMAN-JUDGE
AGREEMENT

100%

HUMAN-VERIFIED
GOLDEN SET (250
ROWS)

40.0%

CORPORATE SAFETY
ESCALATION RATE

D1 — Reconstructing Complete Dialogue Threads with Graph Search

DATA ARCHITECTURE

THE PROBLEM	Raw tweets in twcs.csv are broken into fragmented, out-of-order pieces. Trying to classify an isolated opening tweet without seeing what happened next leads to massive confusion.
THE DECISION	Built a breadth-first search (BFS) graph algorithm to stitch together 12,443 complete, chronological passenger-agent conversation threads.
THE PROOF	Evaluating intents on reconstructed threads improved Macro-F1 by +6.0%, because downstream agent replies clarify ambiguous opening tweets.
RULED OUT	Classifying single tweets in isolation, which leaves out crucial follow-up context.

D2 — 100% Hand-Labelled Golden Test Benchmark (250 Rows)

BENCHMARK INTEGRITY

THE PROBLEM	Many projects use LLMs to label their own test set ("proxy gold"). This causes self-grading bias and artificial score inflation.
THE DECISION	Created and manually hand-annotated a dedicated 250-row human golden set (10 balanced classes × 25 rows) with strict candidate immutability.
THE PROOF	Achieved 100% human review across all 250 rows with zero missing values and zero synthetic label contamination.
RULED OUT	Claiming "high accuracy" using synthetic or model-bootstrapped test labels.

D3 — Stratified 3-Tier Sampling over Naive Random Selection

SAMPLING METHODOLOGY

THE PROBLEM	Random uniform sampling over-samples easy high-frequency complaints and ignores rare, dangerous airline crises.
THE DECISION	Divided the golden set into three distinct tiers: (1) 100 Representative samples, (2) 100 Low-Confidence Boundary samples, and (3) 50 Hard Adversarial samples (sarcasm, hostile rants, multi-issue queries).
THE PROOF	Exposed critical system weaknesses (like sarcasm and gate disputes) during development rather than in production.
RULED OUT	Blind uniform sampling that hides long-tail edge-case failures.

D4 — 100% Rejection of "General Dissatisfaction" Catch-All Bucket

TAXONOMY DESIGN

THE PROBLEM	Unsupervised clustering suggested dumping half of all angry customer tweets into a lazy "General Dissatisfaction" supercluster.
THE DECISION	100% rejected this bucket during human annotation, forcing every complaint into actionable airline categories: Flight Delays, Baggage, Seats, or Disruption.
THE PROOF	0 out of 250 rows accepted the vague candidate label; produced a clean, operationally distinct 10-class airline taxonomy.
RULED OUT	Lazy catch-all sentiment categories that human support agents cannot act on.

D5 — Real-Time Cross-Field Validation Rules in Annotation App

DATA QUALITY

THE PROBLEM	When humans annotate hundreds of rows, fatigue leads to accidental contradictions (e.g. clicking ACCEPT while typing a replacement label).
THE DECISION	Programmed real-time schema validation rules in <code>src/customer_support/labeling/validation.py</code> directly inside the Streamlit labeling tool.
THE PROOF	Prevented human input contradictions before saving, guaranteeing zero schema errors in the final golden set.
RULED OUT	Unchecked free-text spreadsheet labeling that corrupts golden evaluation data.

D6 — Throwing Away 600 Synthetic LLM-Generated Rows

DATA PRUNING

THE PROBLEM	We initially tried generating 600 LLM-labeled synthetic rows to artificially expand our small training dataset.
THE DECISION	Completely discarded all 600 synthetic rows after testing, keeping training strictly restricted to verified human data.
THE PROOF	The 600 synthetic rows gave exactly +0.0 Macro-F1 improvement while adding label noise to fine boundary classes.
RULED OUT	Blindly scaling synthetic data quantity over real human annotation quality.

D7 — Full Conversation Representation for Contextual Feature Learning

FEATURE
ENGINEERING

THE PROBLEM	Live production tweets only contain the customer opener, but historical data contains the full agent exchange.
THE DECISION	Trained the embedding body on full multi-turn conversation text while configuring the inference pipeline to handle customer-only openers.
THE PROOF	Anchored transformer attention on verified historical resolution cues, boosting test accuracy by +4.0%.
RULED OUT	Throwing away historical agent resolution signals during offline model training.

D8 — Hardware Acceleration on Local NVIDIA RTX 4050 Laptop GPU

COMPUTE
INFRASTRUCTURE

THE PROBLEM	Running deep transformer models on CPU was slow, making contrastive parameter tuning painfully sluggish.
THE DECISION	Configured PyTorch CUDA 12.4 to harness the machine's NVIDIA GeForce RTX 4050 Laptop GPU (6GB VRAM) for batched tensor math.
THE PROOF	Encoding speed exceeded 400 sentences/sec; entire end-to-end evaluation harness runs in under 40 seconds.
RULED OUT	Restricting experiments to weak CPU-only models due to compute bottlenecks.

D9 — Rejecting Massive Cross-Entropy Transformer Fine-Tuning

MODEL SELECTION

THE PROBLEM	Standard deep transformers (like RoBERTa-base with 125M parameters) require tens of thousands of samples to avoid severe overfitting.
THE DECISION	Ruled out standard cross-entropy fine-tuning of massive parameter heads on our 200-sample training split.
THE PROOF	RoBERTa cross-entropy collapsed into the majority class (16.0% accuracy), completely failing on few-shot data.
RULED OUT	Brute-force deep neural fine-tuning without few-shot metric learning constraints.

D10 — Contrastive Metric Learning via SetFit on Airline Tweets

FEW-SHOT LEARNING

THE PROBLEM	Pre-trained out-of-the-box sentence embeddings scored only 56.0% accuracy because general text models do not understand airline slang.
THE DECISION	Fine-tuned the MiniLM transformer backbone using SetFit contrastive cosine loss (pulling similar airline queries together, pushing others away).
THE PROOF	Lifted baseline accuracy from 56.0% to 66.0% (+10.0% gain) without requiring thousands of labeled samples.
RULED OUT	Relying on frozen, off-the-shelf sentence embeddings without airline domain adaptation.

D11 — Dual-Backbone Ensemble: Domain Contrastive + Foundation Understanding

ENSEMBLE SOTA
(74%)

THE PROBLEM	Lightweight MiniLM (384d) is fast but struggles with complex, sarcastic, or multi-sentence customer rants.
THE DECISION	Created a dual-backbone ensemble combining domain-adapted SetFit-MiniLM with high-capacity BGE-Large (1024d) foundation embeddings.
THE PROOF	Test accuracy leapt to 74.0% (+18.0% over baseline), with Baggage issues and Flight delays reaching 100% recall.
RULED OUT	Betting on a single small model architecture for complex conversational Twitter text.

D12 — Prior Odds Margin Calibration for Low-Frequency Classes

WINNING SOTA (76%)

THE PROBLEM	Training data had a 4:1 imbalance between Positive experience and Social acknowledgement, causing casual "Hi" tweets to be misclassified as praise.
THE DECISION	Applied prior odds margin calibration to the blended ensemble margins to give rare classes a fair mathematical chance.
THE PROOF	Doubled recall on Social acknowledgement from 25% to 50%, producing our winning headline score of 76.0% accuracy (0.7179 Macro-F1).
RULED OUT	Raw uncalibrated argmax predictions that favor dominant majority classes.

D13 — Strict Liability Safety Gating with Explicit Reason Codes

ROUTING & POLICY

THE PROBLEM	Airlines operate under strict DOT regulations; an AI hallucinating a flight refund, voucher, or safety advice creates legal liability.
THE DECISION	Built a deterministic policy gate: automatically escalates cancellations, baggage loss, profanity, medical needs, and low-confidence predictions to human agents.
THE PROOF	40.0% of test inquiries escalated to human specialists with stated justifications; zero unauthorized compensation promises made.
RULED OUT	Permissive generative auto-replies that hallucinate booking refunds or flight guarantees.

D14 — Grounded Historical Synthesis instead of Blind Verbatim Copying

RAG SYNTHESIS

THE PROBLEM	Naive RAG systems often copy-paste old historical tweets verbatim, accidentally regurgitating other customers' names and handles.
THE DECISION	Indexed 1,500 verified brand resolutions in a FAISS vector store as grounding facts, synthesizing fresh, personalized responses with DM privacy protection.
THE PROOF	Achieved a perfect 5.0 / 5.0 Grounding Quality Score and 0.7485 Ragas Faithfulness with zero personal data leakage.
RULED OUT	Dumping raw historical tweet text verbatim, or unconstrained hallucinated LLM generation.

D15 — Empirical Human-Judge Agreement Calibration

EVALUATION HARNESS

THE PROBLEM	Automated LLM judges can be overly lenient or misaligned with actual customer expectations unless validated against human ratings.
THE DECISION	Benchmarked automated judge scores against independent human expert audits across Grounding, Relevance, Tone, and Actionability.
THE PROOF	Proved strong positive correlation ($r = 0.628$) and a tiny mean absolute difference of only 0.30 points on a 5-point scale.
RULED OUT	Trusting uncalibrated LLM judge metrics without human agreement proof.

American Airlines AI Customer Support Agent • Final Submission • Reproducible in < 40s via `uv run python run_submission.py`